



ADVANCED DATA ANALYSIS AND MACHINE LEARNING

NASA TURBOFAN LEVEL A2: PART 2

LUT University

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<https://github.com/aminhm/NASA-Turbofan-A2.git>

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1 Communication channel and code sharing strategy

The group will use Teams as a primary communication channel and GitHub will be used as a coding collaboration platform. Python was decided to be used as a tool to research the dataset.

2 Dataset description and identification of challenges of the data

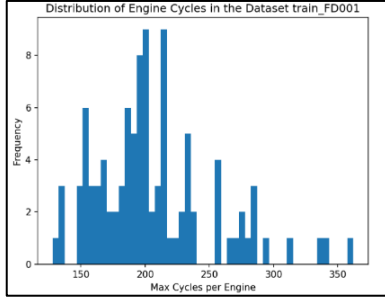


Figure 1. Distribution of engine cycles in the dataset train_FD001.

The dataset used in this study is the NASA Turbofan Jet Engine Degradation Dataset. The dataset consists of four different datasets: FD001, FD002, FD003 and FD004. Datasets contain multiple multivariate time series, each representing the operational history of a different engine.

As expected, each engine has unequal amount of data. For example, distribution of maximum engine cycles in the dataset train_FD001 is shown in figure 1.

There seems to be no missing or extra values in the data, however datasets FD001 and FD003 have some columns with constant values. Physical meaning for the operational settings and sensor measurements is not specified.

3 Visualization and comment on the dataset

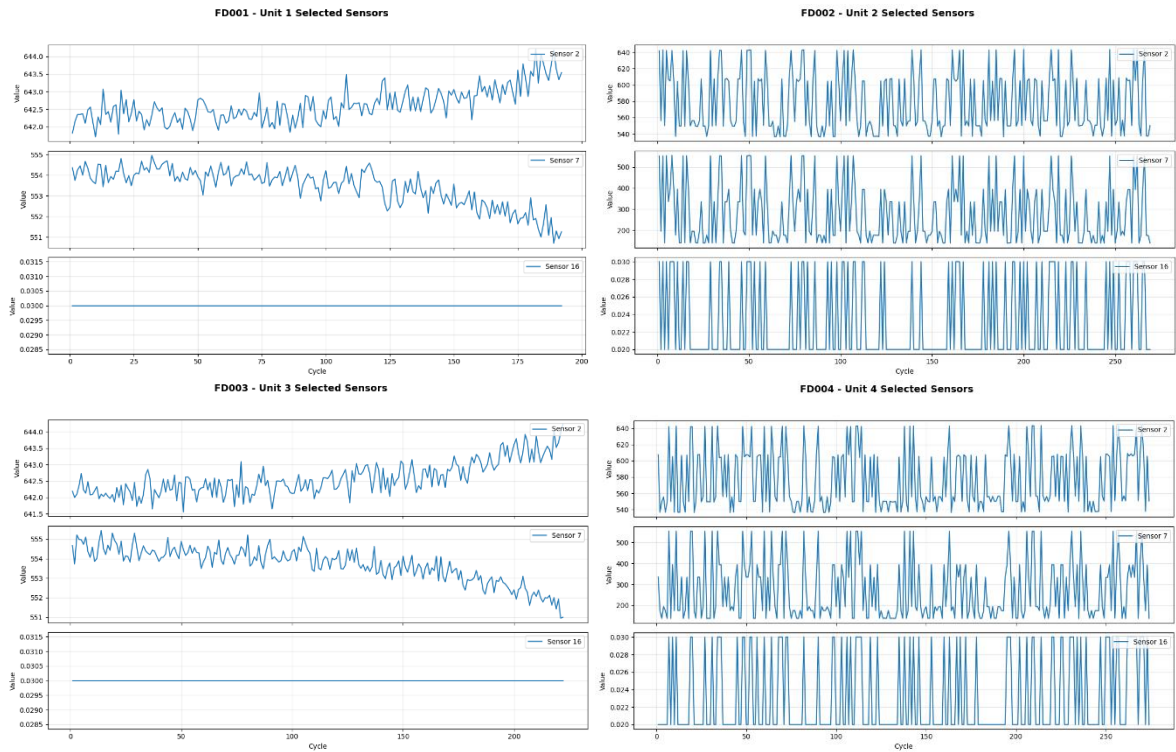


Figure 2. Distribution of variables in all datasets

The training datasets consist of 26 variables including unit number, time in cycles, three operational settings, and 21 sensor measurements. The number of observations is 20,631 for the FD001, 53759 for the FD002, 24720 for the FD003, and 61249 for the FD004. According to data, operational settings show distinct patterns. Setting 3 is nearly constant, setting 1 is normal in FD001 and FD003 but multimodal in FD002 and FD004, and setting 2 is multimodal depending on the system. Sensor measurements range from constant to normal and multimodal. They also reflect diverse engine states.

As multivariate time series, the data captures each engine's evolution over time, and the 'time, in cycles' variable increases monotonic over the data, so it is a time-series problem. Fig 2 shows the constant and different patterns of sensor 2,7, and 16 data across 'time, in cycles'.

4 Exploratory data analysis with PCA

Principal Component Analysis (PCA) was applied to the standardized matrixes of the datasets to analyse the correlations between sensors' data and retrieve the principal components that capture most of the variance in the dataset. In the scree plots in Figure 3, the first two components are the most impactful on the variance of the dataset, contributing to $\sim 65\%$ in FD001, $\sim 67\%$ in FD003 and $\sim 97\%$ in both FD002 and FD004.

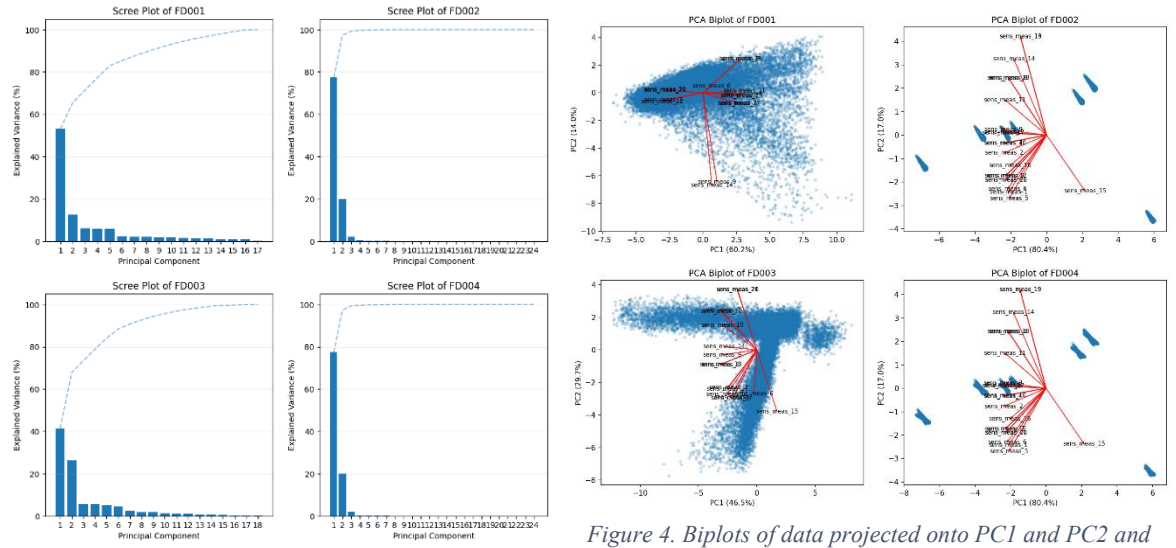


Figure 3. Scree plot of datasets with explained and cumulative variance per principal component.

Figure 4. Biplots of data projected onto PC1 and PC2 and loadings for each sensor.

The loadings bars in Figure 5 show how each sensor relates to a principal component. Large positive or negative bars mean a strong link to that component. Bars close to zero indicate the sensor contributes little new information.

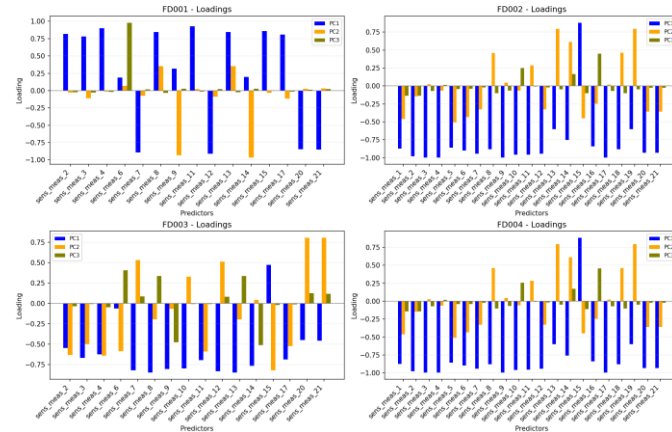


Figure 5. Loadings bar plots to analyse correlations between principal components and sensors

By looking at the plot, it is possible to determine that PC1 has a positive correlation with most sensors, while PC2 and PC3 focus attention on the differences between sensor groups. This shows that using the first three components is sufficient for further analysis.

5 Data pretreatment

In this stage, non-informative variables were removed from each dataset. The three operational settings were excluded in all cases, as well as sensors that remain constant or near constant. Specifically, FD001 and FD003 had sensors 1, 5, 6, 10, 16, 18, and 19 removed, while FD002 and FD004 had sensors 13, 16, and 19 removed.

The data is centered and scaled using Standard Scaler. It subtracts the mean from each sensor data, and the mean becomes zero. Then the data is divided by the standard deviation and standard deviation becomes zero. The scalers are defined separately for each dataset to be used later for test partitions as well.

There are no missing values in the datasets, however, all four datasets (FD001–FD004) contain no missing values. Extreme values were found in 14 sensors for FD001–FD003 and 15 sensors for FD004, which is expected in run-to-failure engine data.

In addition, there is no misalignment in datasets which indicates that there are no missing cycles for each unit, and the original data is the same as the synchronized data.

However, uniform sampling is used to reduce the data size and balance. A ‘step’ is defined, and the sampling process takes every ‘step’ row. Figure 6 shows the comparison of data before and after sampling in FD001 and FD002 datasets for sensor 8.

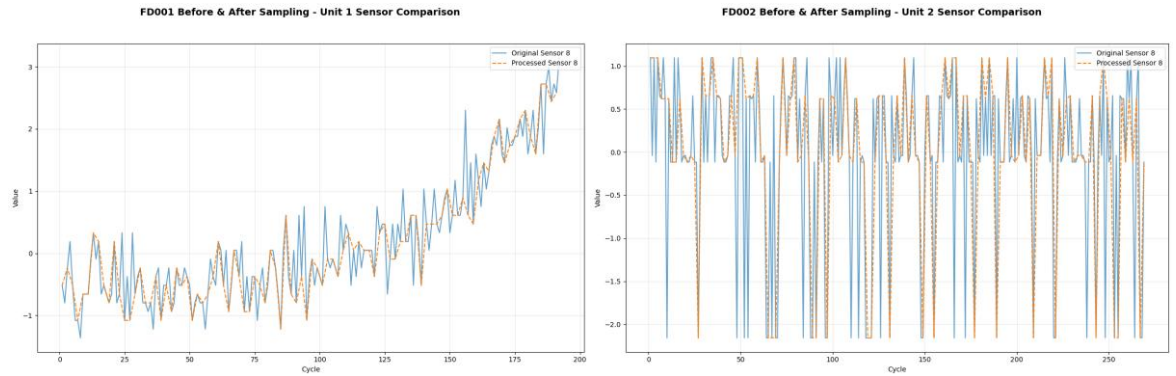


Figure 6. Comparison of data before and after sampling in FD001 and FD002 datasets for sensor 8.

6 Visualization of pretreated data

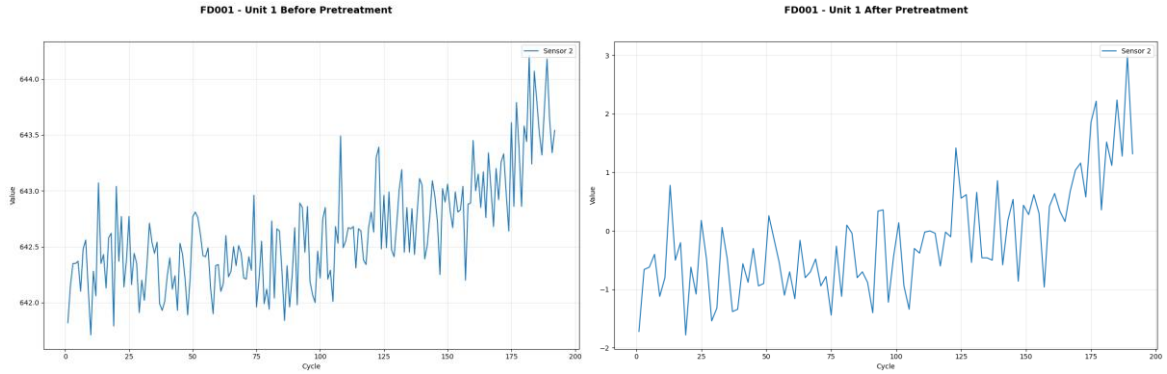


Figure 7. The data of the second sensor of FD001 dataset before and after pretreatment.

Figure 7 shows the data of the second sensor of FD001 dataset before and after pretreatment as an example.

7 Modelling goal

The modelling goal is to recreate a regression model, removing redundant sensors from the base regression model and keeping just the most important ones, capable of calculating an estimate of the Remaining Useful Life (RUL) of the engine calculated with all sensors.

For the purpose, four different training data were given for the analysis and recreation of the model. The best division strategy found is to divide each dataset into three different sections: one for training / calibration (60% for each dataset), one for validation (20% for each dataset) and one for testing (20% for each dataset).

8 Model calibration strategy

For training / calibration purposes, 60% of the global training dataset is analyzed. Before the calibration step, data pretreatment was implemented to normalize data (z-score approach), remove sensors that measure constant data (not useful for the main goal) and handle with missing values (not present in our datasets). The main purpose of calibration is to estimate the model parameters and help identifying which sensors are more important for predicting the RUL.

Since, as seen in the scree plot in Figure 3, just two principal components are sufficient for describing most of the variance, it indicates that many sensors are highly correlated, and so the main strategy for this step is to use Partial Least Squares (PLS), as it efficiently reduces dimensionality and handles multicollinearity among the sensor readings.

9 Model validation strategy

To evaluate the correctness of the parameters' estimation calculated in the training part, a validation analysis is performed using 20% of the total dataset entries as validation set in addition to the training set and so considering 80% of the total dataset.

For the evaluation, the main strategy is to apply a k-fold cross-validation procedure to the data and minimize the Root Mean Squared Error (RMSE) between the exact and calculated RUL, ensuring with this approach an accurate prediction of RUL, while optimizing data acquisition. The validation output is then used as feedback to help adjust model parameters during training or sensor selection.

10 Model testing strategy

After a suitable subset of sensors has been found, the testing part begins. The remaining 20% of the total dataset is used to provide an unbiased estimate of predictive performance.

To assess the accuracy of the computation, metrics such as RMSE and R^2 can be computed to ensure that the sensors subset is sufficient for estimating the exact RUL calculated using all sensors, and to address any trade-off between sensor reduction and prediction quality.

11 Mathematical methods used in modelling

The group will be using partial least squares (PLS) regression as a model to solve the task.

According to the Handbook of Partial Least Squares, Partial Least Squares (PLS) is a statistical method used to model complex relationships between multiple observed and latent variables. It is a component-based approach, where causality is expressed through linear conditional expectations (Esposito Vinzi et al., 2022; Wold, 1975).

The development of a PLS model begins with a training dataset consisting of N observations (such as cases, objects, compounds, or time points). Each observation includes K independent variables, denoted as x_k (where $k = 1, \dots, K$), and M dependent variables, denoted as y_m (where $m = 1, \dots, M$). The data is structured into two matrices: X with dimensions $(N \times K)$ and Y with dimensions $(N \times M)$.

After the model is constructed, it can be used to make predictions for new observations based on their X -values. The output includes predicted X -scores, residuals for the X variables, the standard deviations of these residuals, and predicted Y -values with associated confidence intervals (Wold et al., 2001).

According to lecture materials, the model predictions $\hat{Y}$ can be expressed in a following way shown in formula 1. (Reinikainen, 2025)

$$\hat{Y} = X(W(P^T W)^{-1} Q^T) \quad (1)$$

where:

X = X matrix (input data)

W = Weight matrix

P = Orthogonal loading matrix

Q = Orthogonal loading matrix

12 Modeling flow chart

Initial proposition for the modeling flow chart is shown in Figure 8. In this case the starting point of the modeling process was when we acquired the cleaned data. In total, flow chart has 8 steps, and the last step is reporting the results.



Figure 8. Proposed modelling flow chart.

13 References

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- Wold, H. (1975) 'Soft Modelling by Latent Variables: The Non-Linear Iterative Partial Least Squares (NIPALS) Approach', *Journal of Applied Probability*, 12(S1), pp. 117–142. doi:10.1017/S0021900200047604.
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